Clean Energy

What Is Solar Energy?



The Sun

Our star, 93 million miles away!

Energy = Light

The Sun sends us light and heat energy every single day for FREE!

Huge Amount!

In just 1 hour, the Sun sends enough energy to power all humans for an ENTIRE YEAR!

Never Runs Out

Solar energy is renewable. The Sun will keep shining for billions of years!

How Do Solar Panels Work?

1



Sunlight Hits Panel

Solar panels sit in sunlight, on roofs, in fields, even in space!

2



Cells Get Excited

Tiny silicon cells inside the panel absorb the light energy.

3



Electricity Created

Electrons start moving. This movement IS electricity (DC power)!

4



Inverter Converts It

A device changes DC electricity to AC, the kind homes use.

Where Do We Use Solar Power?



Homes

Rooftop panels power lights, TVs, fridges & more!



Schools

Many schools run on solar to save money for learning!



Electric Cars

Solar charges EV batteries, you can drive on sunshine!



Space Stations

The ISS gets ALL its power from giant solar wings!



Farms

Solar pumps water and powers tools on farms.



Batteries

Solar charges batteries to use power at night!

🤔 Challenges of Solar Power

Solar power is awesome, but it's not perfect yet! Here's what scientists are working on:

Cloudy Days & Night



Solar panels only make power when the Sun is shining. No Sun = no fresh power!

Upfront Cost



Solar panels can be expensive to buy and install, though they save money over time!

Needs Lots of Space



Big solar farms need a LOT of land. That's space other things could use!

Panel Recycling



Old solar panels are hard to recycle. Scientists are working on better solutions!

How Are We Improving Solar?



Better Batteries

New batteries store extra solar power so we can use it at night or when it's cloudy.



Solar Windows

Scientists made special glass that makes electricity, windows that are also solar panels!



Flexible Solar Panels

Thin, bendy solar panels can go on backpacks, clothes, and curved surfaces!



AI Tracking

Smart computers turn panels to follow the Sun all day, making 30% more power!

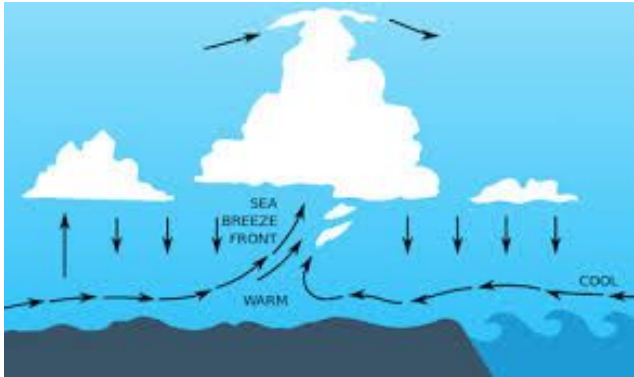
Wind

Wind is the movement of air from high pressure to low pressure areas

The sun heats Earth's surface unevenly; land heats faster than water, equator more than poles

These temperature differences cause air to rise and fall, creating pressure gaps

Wind is nature's way of balancing pressure differences across the atmosphere



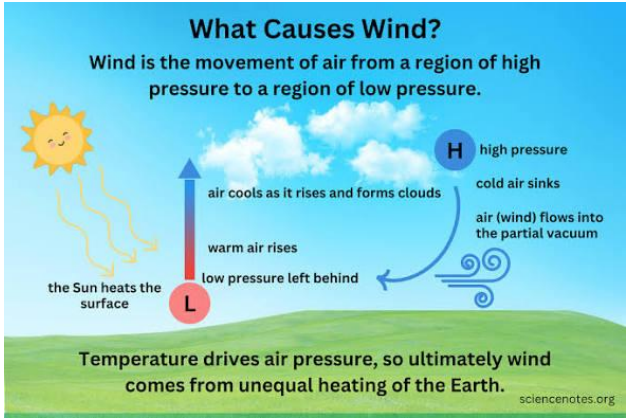
How is wind created?

The sun heats the ground, warming the air above it

Warm air becomes lighter and rises

Cooler, denser air rushes in to fill the gap: that movement is wind

This happens at every scale, from a beach breeze to global wind patterns



Why does wind curve?

Wind doesn't blow in a straight line; it curves due to the Coriolis Effect

The Coriolis Effect is caused by Earth's rotation

In the Northern Hemisphere, wind deflects to the right; in the Southern, to the left

This is why hurricanes spin counterclockwise in the north and clockwise in the south



Wind Energy

Why does wind matter for Earth day? WE CAN USE IT TO MAKE ENERGY!

When wind blows, it spins the blades, this helps create electricity

Best part?

Doesn’t pollute air! A single modern turbine can power hundreds of homes

Land around turbines can still be used for farming

Denmark generates over 50% of its electricity wind

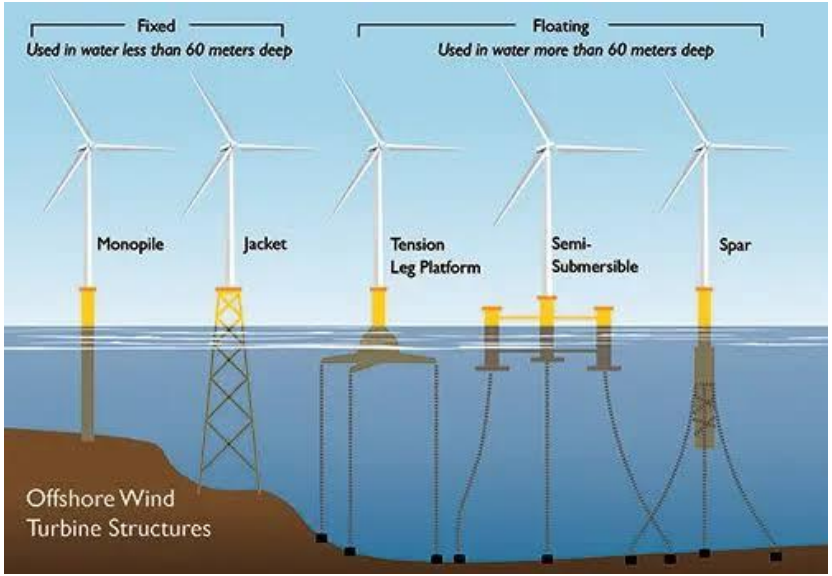


Challenges and Solutions

What happens if the wind stops blowing? Do we still get energy?

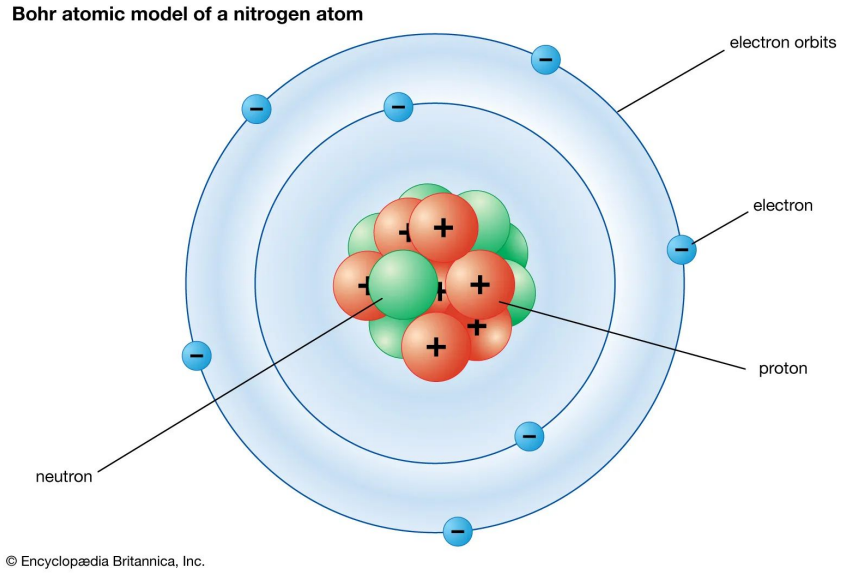
That's one of the biggest problems.

Scientists are working on solutions.



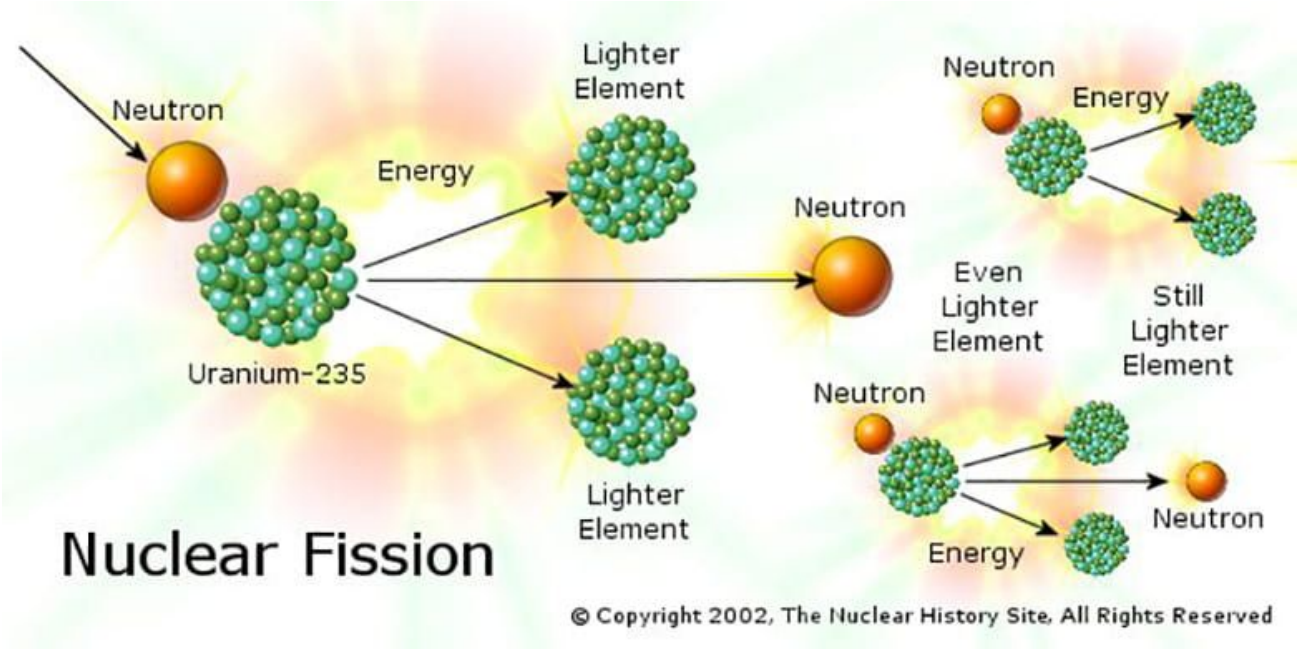
Nuclear

Atomic Interactions:

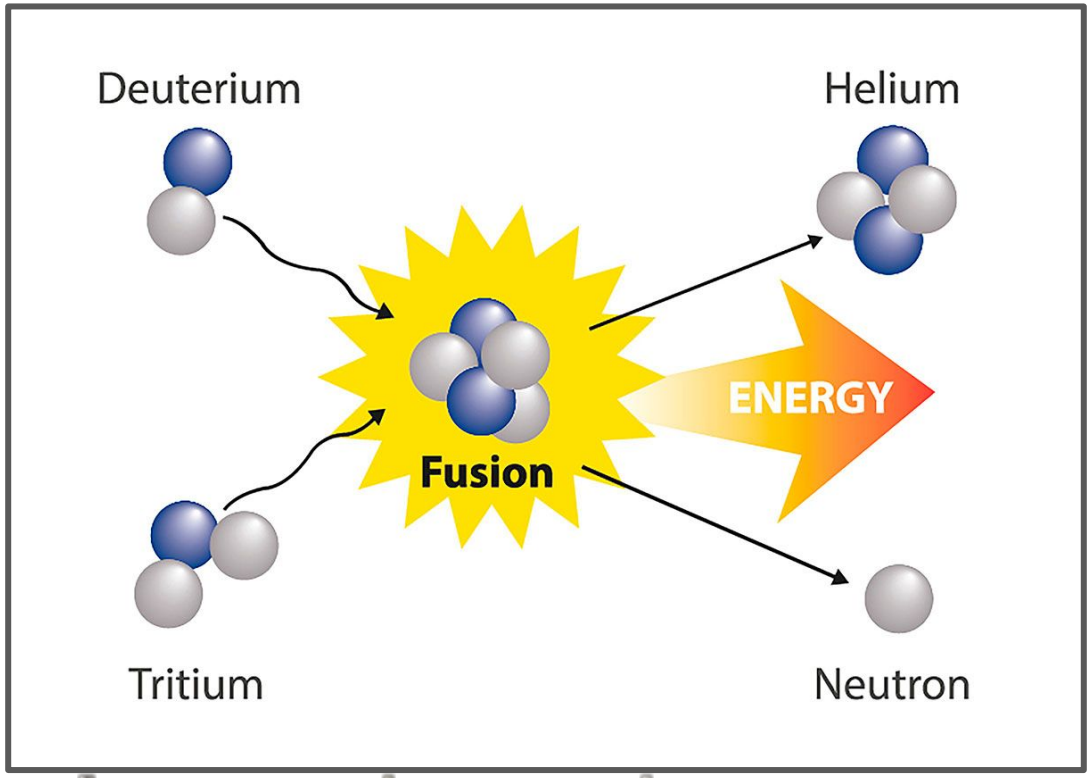


$$^{10}_{5}B + ^{1}_{0}n \rightarrow ^{7}_{3}Li + ^{4}_{2}He + 2.8\,Mev$$

Fission


Neutron
Energy
Lighter Element
Uranium-235
Neutron
Lighter Element
Neutron
Energy
Even Lighter Element
Still Lighter Element
Neutron
Neutron
Energy
Nuclear Fission
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Fusion



$$ {}^{2}_{1}H + {}^{3}_{1}H \rightarrow {}^{4}_{2}He + {}^{1}_{0}n + 17.59MeV $$

Benefits/Drawbacks to Nuclear

Reliability

Nuclear reactors are able to operate at 93% efficiency



Energy Density

A small pellet of uranium produces the same energy as a ton of coal

Nuclear Waste

Used nuclear fuel is still radioactive and can be very harmful

Money

Costs a lot of money to build